

Standard calculation procedures and examples of calculation results for 4-quantization-level, 8-neighbor GLCMs. (By H.Nishiyama / Niigata Univ. 2026/07/05, Ver 2.2)

In PyRadiomics, the angle-related symmetrization parameter 'weightingNorm' is set to 'None' by default. GLCMs are generated for each angle to calculate feature values, which are ultimately averaged.

The GLCM features obtained using this procedure vary depending on the rotation angle of the target image.

<https://github.com/AIM-Harvard/pyradiomics>

[https://github.com/auinen/pyradiomics/blob/master/workspace/check\\_P\\_glcml.ipynb](https://github.com/auinen/pyradiomics/blob/master/workspace/check_P_glcml.ipynb)

To obtain rotation-invariant GLCM features for the target image, you must set 'weightingNorm' to 'no\_weighting'.

[https://github.com/auinen/pyradiomics/blob/master/workspace/radiomics\\_julia\\_2D.ipynb](https://github.com/auinen/pyradiomics/blob/master/workspace/radiomics_julia_2D.ipynb)

| Final Result            |         | Results near GLCM-8<br>(simplified version) and<br>differences |             |
|-------------------------|---------|----------------------------------------------------------------|-------------|
| Average of 4 directions |         | values                                                         | differences |
| Joint maximum           | 0.18112 | 0.15714                                                        | 0.0240      |
| Joint average           | 2.01818 | 2.01667                                                        | 0.0015      |
| Joint variance          | 0.96322 | 0.96401                                                        | -0.0008     |
| Joint entropy           | 3.53613 | 3.60614                                                        | -0.0700     |

| 8-neighborhood GLCM<br>(normalized) |       |       |       |
|-------------------------------------|-------|-------|-------|
| 0.119                               | 0.143 | 0.062 | 0.031 |
| 0.143                               | 0.157 | 0.045 | 0.048 |
| 0.062                               | 0.045 | 0.010 | 0.017 |
| 0.031                               | 0.048 | 0.017 | 0.024 |

| Original Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | angular direction |       | 8-neighbor GLCM in the angular direction<br>(before normalization) |       |   |   | 8-neighbor GLCM in the angular direction<br>(after normalization) |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                   |       |                                                                    |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| <table><tr><td>1</td><td>2</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>3</td><td>4</td><td>3</td><td>1</td><td>2</td><td>2</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td><td>2</td><td>4</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>2</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td></tr><tr><td>1</td><td>3</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>2</td><td>4</td><td>2</td><td>2</td><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td><td>2</td><td>4</td><td>1</td><td>3</td><td>3</td><td>3</td></tr><tr><td>3</td><td>2</td><td>1</td><td>4</td><td>4</td><td>1</td><td>1</td><td>1</td></tr></table> | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 | 1 | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 2 | 4 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 4 | 1 | 3 | 3 | 3 | 3 | 2 | 1 | 4 | 4 | 1 | 1 | 1 | 0 — 180   |  | <table><tr><td>16</td><td>14</td><td>5</td><td>4</td></tr><tr><td>14</td><td>22</td><td>3</td><td>6</td></tr><tr><td>5</td><td>3</td><td>4</td><td>2</td></tr><tr><td>4</td><td>6</td><td>2</td><td>2</td></tr></table> | 16 | 14 | 5 | 4 | 14 | 22 | 3 | 6 | 5 | 3 | 4 | 2 | 4 | 6 | 2 | 2 | <table><tr><td>1</td><td>0.143</td><td>0.125</td><td>0.045</td><td>0.036</td></tr><tr><td>2</td><td>0.125</td><td>0.196</td><td>0.027</td><td>0.054</td></tr><tr><td>3</td><td>0.045</td><td>0.027</td><td>0.036</td><td>0.018</td></tr><tr><td>4</td><td>0.036</td><td>0.054</td><td>0.018</td><td>0.018</td></tr></table> |  |  |  | 1 | 0.143 | 0.125 | 0.045 | 0.036 | 2 | 0.125 | 0.196 | 0.027 | 0.054 | 3 | 0.045 | 0.027 | 0.036 | 0.018 | 4 | 0.036 | 0.054 | 0.018 | 0.018 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 1     | 3                                                                  | 4     | 3 | 1 | 2                                                                 | 2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                 | 2     | 4                                                                  | 2     | 2 | 4 | 1                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                 | 2     | 1                                                                  | 1     | 2 | 3 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 2     | 2                                                                  | 1     | 2 | 2 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 4     | 2                                                                  | 2     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 2     | 4                                                                  | 1     | 3 | 3 | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 1     | 4                                                                  | 4     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 14                | 5     | 4                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 22                | 3     | 6                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 4     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6                 | 2     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.143             | 0.125 | 0.045                                                              | 0.036 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.125             | 0.196 | 0.027                                                              | 0.054 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.045             | 0.027 | 0.036                                                              | 0.018 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.036             | 0.054 | 0.018                                                              | 0.018 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| <table><tr><td>1</td><td>2</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>3</td><td>4</td><td>3</td><td>1</td><td>2</td><td>2</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td><td>2</td><td>4</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>2</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td></tr><tr><td>1</td><td>3</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>2</td><td>4</td><td>2</td><td>2</td><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td><td>2</td><td>4</td><td>1</td><td>3</td><td>3</td><td>3</td></tr><tr><td>3</td><td>2</td><td>1</td><td>4</td><td>4</td><td>1</td><td>1</td><td>1</td></tr></table> | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 | 1 | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 2 | 4 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 4 | 1 | 3 | 3 | 3 | 3 | 2 | 1 | 4 | 4 | 1 | 1 | 1 | 45 — 225  |  | <table><tr><td>6</td><td>16</td><td>6</td><td>5</td></tr><tr><td>16</td><td>12</td><td>7</td><td>4</td></tr><tr><td>6</td><td>7</td><td>0</td><td>1</td></tr><tr><td>5</td><td>4</td><td>1</td><td>2</td></tr></table>  | 6  | 16 | 6 | 5 | 16 | 12 | 7 | 4 | 6 | 7 | 0 | 1 | 5 | 4 | 1 | 2 | <table><tr><td>1</td><td>0.061</td><td>0.163</td><td>0.061</td><td>0.051</td></tr><tr><td>2</td><td>0.163</td><td>0.122</td><td>0.071</td><td>0.041</td></tr><tr><td>3</td><td>0.061</td><td>0.071</td><td>0.000</td><td>0.010</td></tr><tr><td>4</td><td>0.051</td><td>0.041</td><td>0.010</td><td>0.020</td></tr></table> |  |  |  | 1 | 0.061 | 0.163 | 0.061 | 0.051 | 2 | 0.163 | 0.122 | 0.071 | 0.041 | 3 | 0.061 | 0.071 | 0.000 | 0.010 | 4 | 0.051 | 0.041 | 0.010 | 0.020 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 1     | 3                                                                  | 4     | 3 | 1 | 2                                                                 | 2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                 | 2     | 4                                                                  | 2     | 2 | 4 | 1                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                 | 2     | 1                                                                  | 1     | 2 | 3 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 2     | 2                                                                  | 1     | 2 | 2 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 4     | 2                                                                  | 2     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 2     | 4                                                                  | 1     | 3 | 3 | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 1     | 4                                                                  | 4     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 16                | 6     | 5                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 12                | 7     | 4                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 7                 | 0     | 1                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4                 | 1     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.061             | 0.163 | 0.061                                                              | 0.051 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.163             | 0.122 | 0.071                                                              | 0.041 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.061             | 0.071 | 0.000                                                              | 0.010 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.051             | 0.041 | 0.010                                                              | 0.020 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| <table><tr><td>1</td><td>2</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>3</td><td>4</td><td>3</td><td>1</td><td>2</td><td>2</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td><td>2</td><td>4</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>2</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td></tr><tr><td>1</td><td>3</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>2</td><td>4</td><td>2</td><td>2</td><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td><td>2</td><td>4</td><td>1</td><td>3</td><td>3</td><td>3</td></tr><tr><td>3</td><td>2</td><td>1</td><td>4</td><td>4</td><td>1</td><td>1</td><td>1</td></tr></table> | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 | 1 | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 2 | 4 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 4 | 1 | 3 | 3 | 3 | 3 | 2 | 1 | 4 | 4 | 1 | 1 | 1 | 90 — 270  |  | <table><tr><td>14</td><td>18</td><td>9</td><td>2</td></tr><tr><td>18</td><td>12</td><td>5</td><td>7</td></tr><tr><td>9</td><td>5</td><td>0</td><td>1</td></tr><tr><td>2</td><td>7</td><td>1</td><td>2</td></tr></table> | 14 | 18 | 9 | 2 | 18 | 12 | 5 | 7 | 9 | 5 | 0 | 1 | 2 | 7 | 1 | 2 | <table><tr><td>1</td><td>0.125</td><td>0.161</td><td>0.080</td><td>0.018</td></tr><tr><td>2</td><td>0.161</td><td>0.107</td><td>0.045</td><td>0.063</td></tr><tr><td>3</td><td>0.080</td><td>0.045</td><td>0.000</td><td>0.009</td></tr><tr><td>4</td><td>0.018</td><td>0.063</td><td>0.009</td><td>0.018</td></tr></table> |  |  |  | 1 | 0.125 | 0.161 | 0.080 | 0.018 | 2 | 0.161 | 0.107 | 0.045 | 0.063 | 3 | 0.080 | 0.045 | 0.000 | 0.009 | 4 | 0.018 | 0.063 | 0.009 | 0.018 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 1     | 3                                                                  | 4     | 3 | 1 | 2                                                                 | 2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                 | 2     | 4                                                                  | 2     | 2 | 4 | 1                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                 | 2     | 1                                                                  | 1     | 2 | 3 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 2     | 2                                                                  | 1     | 2 | 2 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 4     | 2                                                                  | 2     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 2     | 4                                                                  | 1     | 3 | 3 | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 1     | 4                                                                  | 4     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 18                | 9     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 12                | 5     | 7                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5                 | 0     | 1                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 7                 | 1     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.125             | 0.161 | 0.080                                                              | 0.018 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.161             | 0.107 | 0.045                                                              | 0.063 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.080             | 0.045 | 0.000                                                              | 0.009 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.018             | 0.063 | 0.009                                                              | 0.018 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| <table><tr><td>1</td><td>2</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>3</td><td>4</td><td>3</td><td>1</td><td>2</td><td>2</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td><td>2</td><td>4</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>2</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td></tr><tr><td>1</td><td>3</td><td>2</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td></tr><tr><td>1</td><td>2</td><td>4</td><td>2</td><td>2</td><td>1</td><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td><td>2</td><td>4</td><td>1</td><td>3</td><td>3</td><td>3</td></tr><tr><td>3</td><td>2</td><td>1</td><td>4</td><td>4</td><td>1</td><td>1</td><td>1</td></tr></table> | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 | 1 | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 3 | 1 | 1 | 3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 2 | 4 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 4 | 1 | 3 | 3 | 3 | 3 | 2 | 1 | 4 | 4 | 1 | 1 | 1 | 135 — 315 |  | <table><tr><td>14</td><td>12</td><td>6</td><td>2</td></tr><tr><td>12</td><td>20</td><td>4</td><td>3</td></tr><tr><td>6</td><td>4</td><td>0</td><td>3</td></tr><tr><td>2</td><td>3</td><td>3</td><td>4</td></tr></table> | 14 | 12 | 6 | 2 | 12 | 20 | 4 | 3 | 6 | 4 | 0 | 3 | 2 | 3 | 3 | 4 | <table><tr><td>1</td><td>0.143</td><td>0.122</td><td>0.061</td><td>0.020</td></tr><tr><td>2</td><td>0.122</td><td>0.204</td><td>0.041</td><td>0.031</td></tr><tr><td>3</td><td>0.061</td><td>0.041</td><td>0.000</td><td>0.031</td></tr><tr><td>4</td><td>0.020</td><td>0.031</td><td>0.031</td><td>0.041</td></tr></table> |  |  |  | 1 | 0.143 | 0.122 | 0.061 | 0.020 | 2 | 0.122 | 0.204 | 0.041 | 0.031 | 3 | 0.061 | 0.041 | 0.000 | 0.031 | 4 | 0.020 | 0.031 | 0.031 | 0.041 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 2     | 2                                                                  | 2     | 1 | 2 | 2                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                 | 1     | 3                                                                  | 4     | 3 | 1 | 2                                                                 | 2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                 | 2     | 4                                                                  | 2     | 2 | 4 | 1                                                                 | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                 | 2     | 1                                                                  | 1     | 2 | 3 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 2     | 2                                                                  | 1     | 2 | 2 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 4     | 2                                                                  | 2     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 2     | 4                                                                  | 1     | 3 | 3 | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                 | 1     | 4                                                                  | 4     | 1 | 1 | 1                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 12                | 6     | 2                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 20                | 4     | 3                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4                 | 0     | 3                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                 | 3     | 4                                                                  |       |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.143             | 0.122 | 0.061                                                              | 0.020 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.122             | 0.204 | 0.041                                                              | 0.031 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.061             | 0.041 | 0.000                                                              | 0.031 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.020             | 0.031 | 0.031                                                              | 0.041 |   |   |                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |  |                                                                                                                                                                                                                         |    |    |   |   |    |    |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                             |  |  |  |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |   |       |       |       |       |

'weightingNorm' = None

|                |         |
|----------------|---------|
| Joint maximum  | 0.19643 |
| Joint average  | 2.02679 |
| Joint variance | 0.97250 |
| Joint entropy  | 3.57109 |

|                |         |
|----------------|---------|
| Joint maximum  | 0.16327 |
| Joint average  | 2.05102 |
| Joint variance | 0.96678 |
| Joint entropy  | 3.57317 |

|                |         |
|----------------|---------|
| Joint maximum  | 0.16071 |
| Joint average  | 1.96429 |
| Joint variance | 0.94515 |
| Joint entropy  | 3.48574 |

|                |         |
|----------------|---------|
| Joint maximum  | 0.20408 |
| Joint average  | 2.03061 |
| Joint variance | 0.96845 |
| Joint entropy  | 3.51451 |

rotation-variant GLCM  
features for the target  
image

rotation-invariant GLCM  
features for the target  
image

| 8-neighborhood GLCM<br>(before normalization) |    |    |    |
|-----------------------------------------------|----|----|----|
| 50                                            | 60 | 26 | 13 |
| 60                                            | 66 | 19 | 20 |
| 26                                            | 19 | 4  | 7  |
| 13                                            | 20 | 7  | 10 |

'weightingNorm' = 'no\_weighting'